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STAFF REPORT ON THE 2014 ARTICLE IV CONSULTATION, THE THIRD REVIEW UNDER THE STANDBY CREDIT FACILTY ARRANGEMENT, REQUEST FOR A WAIVER FOR NONOBSERVANCE OF A PERFORMANCE CRITERION, AND FINANCING ASSURANCES REVIEW—DEBT SUSTAINABILITY ANALYSIS

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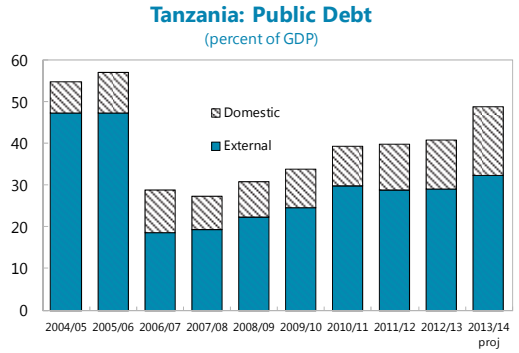
Risk of external debt distress:	Low
Augmented by significant risks stemming from domestic public and/or private external debt?	No

Tanzania continues to face a low risk of debt distress based on the external debt sustainability analysis (DSA). The external debt outlook improves as a result of an increase in the discount rate, but this is partly offset by (i) a slower pace of fiscal deficit reduction and (ii) a shift in the long-term composition of external financing towards more non-concessional loans. While the baseline public debt outlook remains favorable, continued fiscal consolidation is critical. In addition, recognizing the outstanding pension and other liabilities has an impact on the level of public debt. The baseline numerical analysis in this DSA does not take into account prospects related to natural gas resources whose scale, while potentially favorable, remains uncertain. These results highlight the need for Tanzania to sustain fiscal consolidation efforts, to adopt a conservative approach to non-concessional borrowing, to promptly address pension-related liabilities, and to further improve its debt management capacity.[1]

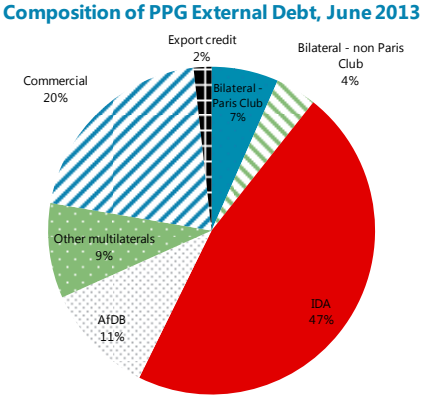
[1] This DSA replaces the previous one prepared in June 2012. Tables and figures are in fiscal years (July-June). For example, 2013 refers to fiscal year 2012/13. Under the Country Policy and Institutional Assessment (CPIA), Tanzania is rated as a strong performer, with an average rating of 3.73 during 2010-12. Although the average CPIA score is slightly below the boundary for strong performance category (3.75), the performance rating remains strong since this is the first year of breaching the boundary and the breach is less than 0.5.

BACKGROUND

1. Tanzania's public and publicly guaranteed (PPG) external debt as a share of GDP has steadily crept upwards in recent years (text figure, left panel). Since the Multilateral Debt Relief Initiative (MDRI) in 2006/07, which reduced the external debt-to-GDP ratio from 47 percent to 19 percent, PPG external debt (excluding arrears under negotiation for relief) has increased to about 29 percent of GDP at end-June 2013, or 17 percent of GDP in present value terms using a 5 percent discount rate[2]. Total public sector debt (external plus domestic) stood at about US$12.4 billion or 40.9 percent of GDP at end-June 2013, rising from a low of 27.5 percent at end-June 2008.



Sources: Ministry of Finance, Bank of Tanzania, and IMF staff calculations.
Note: 2013/14 projection of domestic debt includes a one-off adjustment for recognizing outstanding domestic liabilities, including past-due pension obligations, guarantees for public enterprises, and domestic expenditure arrears.



Sources: Ministry of Finance, Bank of Tanzania, and IMF staff calculations.
Note: Excludes arrears to non-Paris Club creditors under negotiation for relief.

2. The majority of Tanzania's PPG external debt is still concessional, but borrowing on non-concessional terms has increased in recent years (text figure, right panel). At end-June 2013, more than two-thirds of total public external debt was owed to multilateral institutions, of which the International Development Association (IDA) and the African Development Bank (AfDB) constitute the largest creditors. Government borrowing from commercial sources amounted to about 20 percent of the total stock at end-June 2013, having risen rapidly from about 2 percent at end-June 2010. In February 2013, Tanzania raised US$600 million from international capital markets in a private placement transaction.[3] Tanzania is currently in the process of obtaining its sovereign risk rating, which would facilitate future Eurobond issuance.

3. Domestic public debt stood at 12 percent of GDP at end-June 2013, and mostly consisted of Treasury bonds. Domestic public debt is held primarily by commercial banks (39 percent), the Bank of Tanzania (35 percent), and pension funds and insurance companies (23 percent). Private external debt has risen in recent years to about 5 percent of GDP in 2012/13, though there is considerable uncertainty regarding private external debt data.

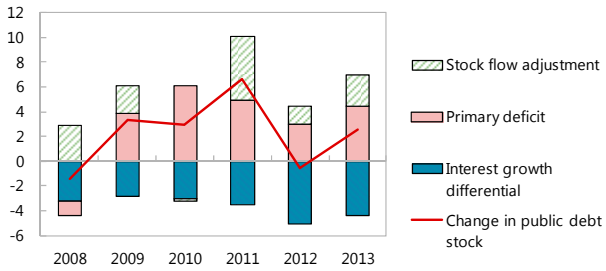
[2] The discount rate was increased to 5 percent as of October 2013. See "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" (SM/13/292).

[3] This is an amortizing bond with a maturity of seven years, and was priced at 600 basis points over 6-month LIBOR at the time of issuance. The amortizing feature helps reduce rollover risks, whereas the floating interest rate on the bond implies increased exposure to changing interest rates.

4. The current DSA only covers central government debt. Owing to data limitations, borrowing by local governments and most public enterprises is excluded, thus potentially understating the total debt level. The external public debt stock includes the foreign currency portion of commercial loans contracted by the state-owned power utility TANESCO[4]. For the purpose of the current DSA, the domestic public debt stock incorporates starting in 2013/14 several outstanding government liabilities that have yet to be formally recognized as debt (but that were included in the DSA published by the MoF), amounting to about Tsh4 trillion or 7 percent of GDP. These include outstanding liabilities to the Public Service Pension Fund (Tsh1160 billion)[5], outstanding unpaid domestic claims (Tsh892 billion)[6], government guarantees for public enterprises (Tsh912 billion), and other actual liabilities arising from past PPP transactions, court orders, and explicit government guarantees.[7]

5. Stock-flow adjustments have accounted for a sizeable portion of the rise in Tanzania's public debt in recent years. A decomposition of annual change in gross public debt shows that stock-flow residuals averaged 2.4 percent of GDP over 2008-13, peaking at 5.2 percent of GDP in 2011. The primary fiscal deficit also made a sizable contribution to the debt ratio increase. The interest-growth differential (i.e., the difference between effective interest rates[8] and output growth rates) was considerably negative, reflecting the still-high concessionality of the debt stock, thereby slowing the pace of the debt ratio increase.

Stock Flow Adjustments
(percent of GDP)



Sources: IMF staff calculations.

6. The authorities continue to make progress in upgrading their debt management capacity. With technical support from the World Bank and the East African MEFMI, the MoF conducted a debt sustainability assessment in September 2013. The authorities' medium-term debt strategy (MTDS) was also updated in November 2013 with support from a joint IMF-World Bank technical assistance mission. Both the DSA and the MTDS report have been published on the MoF's web site, and are expected to serve as key inputs that will guide the government's borrowing policy going forward. There is an ongoing process to put into operation a centralized debt management office (DMO), housed at the MoF, that will bring together the fragmented debt management functions.

[4] TANESCO borrowing in foreign currency amounts to about US$20 million in 2012/13 and US$84 million in 2013/14.

[5] Estimate from the government's verification exercise, and consistent with estimates by the World Bank and the Genesis Report. Contingent liabilities arising from financial weakness of the broader pension sector can be substantially higher.

[6] Unpaid claims of duration above 90 days, estimated as of end-December 2013. Ministries with large claims include Works, Health, Education, Agriculture, Home Affairs, and Defense.

[7] In addition, the government has significant multi-year commitments arising from existing contracts, especially in the road sector, which are not included in the DSA.

[8] Effective interest rates are computed by dividing total interest payments by last period's debt stock.

Selected Macroeconomic Indicators, Current vs. Previous DSA

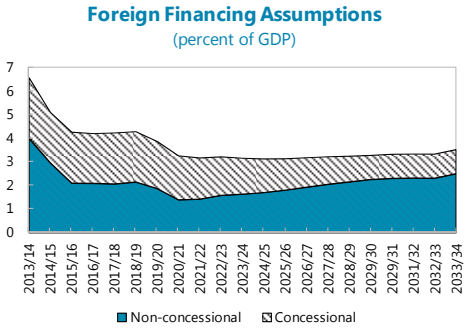
	2014	2015	2016	Long term 1/
Real GDP growth (percent)				
Current DSA	7.1	7.1	7.1	7.0
Previous DSA	6.9	7.1	7.1	7.5
Fiscal balance (% GDP)				
Current DSA	-5.2	-4.9	-4.0	-3.0
Previous DSA	-4.5	-3.9	-3.4	-2.6
Current account balance (% GDP)				
Current DSA	-14.5	-13.8	-12.9	-10.2
Previous DSA	-12.5	-12.2	-11.9	-9.8
Net FDI (% GDP)				
Current DSA	6.0	6.7	6.7	6.1
Previous DSA	6.1	6.0	5.9	5.8

1/ Defined as the last 15 years of the projection period. For the current DSA, the long term covers the period 2020-34, whereas for the previous DSA it covered 2018-32.

UNDERLYING ASSUMPTIONS

7. Compared with the previous DSA, the current assessment is based on a somewhat slower pace of fiscal deficit reduction (text table). This reflects Tanzania's need to increase growth-supporting infrastructure investments while maintaining a sustainable debt outlook. The deficit is projected to narrow gradually to 4 percent of GDP by 2015/16, and to further decline to 3 percent of GDP in 2020/21 and thereafter in line with the convergence criteria agreed under the East African Monetary Union (EAMU) protocol. Domestic revenues (excluding grants) are projected to grow from 18 percent of GDP in 2012/13 to over 23 percent of GDP by 2033/34, consistent with economic development and Tanzania's tax potential. External grants are assumed to decline gradually, from 3.4 percent of GDP in 2013/14 to 2 percent of GDP in the long term.

8. The composition of external borrowing is assumed to gradually shift toward non-concessional debt, consistent with a reduction in aid dependency as the economy matures. External non-concessional borrowing (ENCB) is expected to pick up in 2013/14, due in part to a US$558 million projected disbursement of the gas pipeline loan[9], and in part to debt contracted to finance new infrastructure projects.[10] Beyond 2013/14, the assumption is that total gross foreign financing would consist of about half ENCB and half concessional (program and project) loans in the medium term. Thereafter,



[9] This is part of a US$920 million loan from China EXIM Bank, contracted in 2012/13, to build a new gas pipeline that would bring low-cost natural gas from shallow water offshore reserves to the Dar es Salaam area.

[10] Contracted ENCB is projected at US$700 million in 2013/14, consistent with the second review under the SCF arrangement.

ENCB would gradually become the more prominent financing source, reaching about 70 percent of total foreign financing requirements by 2033/34.

9. Other macroeconomic assumptions remain broadly similar to the previous DSA (see Box 1 for detail). Real GDP is assumed to keep growing by 7 percent annually over the projection period, with continued strong capital accumulation. The external current account (CA) is projected to gradually improve over the medium term as the availability of gas-based power generation reduces the need for oil imports, and to stabilize at a still-large deficit of 10 percent of GDP.

Box 1. Key DSA Assumptions

- **Growth, inflation, exchange rate.** Real GDP growth is projected to remain at 7 percent over the medium and long term. Inflation would converge to the Bank of Tanzania's medium-term target of 5 percent, yielding a nominal growth rate of about 12 percent per annum. The Tanzanian shilling is projected to depreciate against the USD by 3 percent annually in the medium and long run, due to the inflation differential between Tanzania and the US.
- **Exports and imports.** Export and import values are assumed to grow at 11 and 10 percent annually in the long term, respectively.
- **Fiscal deficit.** The fiscal deficit is projected to decline to 4 percent of GDP by 2015/16, and further to 3 percent of GDP from 2020/21 onwards. Revenues excluding grants are assumed to grow from 18 percent of GDP to 23 percent of GDP by 2033/34.
- **Aid and FDI flows.** External grants are assumed to gradually decline to 2 percent of GDP by 2033/34. Foreign concessional loans (program and project loans) would slowly decline to 1 percent of GDP in the long term. Concessional loans are assumed to continue to come primarily from multilateral creditors (IDA, AfDB, etc.) in the medium term, but would then be increasingly provided by bilateral – particularly non-Paris Club – creditors, whose financing terms are not as favorable. After a slight increase due to further offshore natural gas explorations, net FDI inflows would stabilize at an average of 6.1 percent of GDP in the long term.
- **Domestic borrowing.** Net domestic financing is kept at a maximum of about 1 percent of GDP throughout the projection period. New domestic debt is assumed to carry a real interest rate of 10 percent with average maturity of seven years, consistent with recent experience.
- **External non-concessional borrowing.** ENCB is assumed to finance about half of the gross foreign financing requirement up to 2019/20 before gradually gaining importance over concessional sources, reaching 2.5 percent of GDP by 2033/34. The grant element of new borrowing is projected to decline considerably over the long run. New external commercial borrowing is assumed to be provided by a combination of export credit agencies (ECA) and commercial banks (syndicated loans), with the latter source becoming more prominent in the long run. Consistent with prevailing market conditions, ECA loans have an average interest rate of 4 percent with a 15-year maturity, whereas syndicated loans carry a higher average interest rate (7 percent) and shorter maturity (7 years).

10. The current DSA includes a government pension liability in the 2013/14 domestic public debt stock. The government currently has Tsh1.16 trillion (about 2.1 percent of GDP) of past-due payment liabilities it owes to PSPF for pension benefits related to the pre-1999 noncontributory scheme.[11] Although the authorities have acknowledged the outstanding liability and budget for partial repayment each year, the liability has not been formally recorded in the fiscal accounts and thus was not part of the domestic debt stock. This DSA assumes that recognition of this liability (and several other actual liabilities) occurs in 2013/14, resulting in a one-off increase in the 2013/14 public debt stock.[12] Moreover, in interpreting the fiscal deficit assumptions that underlie this DSA (and the related policy effort to meet such assumptions), it is important to keep in mind the flow of future expenditure obligations (estimated at 0.5 percent of GDP per year in the medium term, gradually slowing down thereafter) to cover the pension benefits of pensioners in the pre-1999 system as a part of the deficit.

11. The current DSA incorporates better-refined projections of debt service on existing external debts. The assessment uses individual loan information provided by the authorities, reflecting their process of upgrading the debt database and debt monitoring capacity. Future debt service obligations are projected for each loan, then aggregated into several representative instruments (e.g., by creditor type and lending terms).

12. The discount rate is currently set at 5 percent, compared to 4 percent used in the previous DSA.

EXTERNAL DSA

13. All PPG external debt burden indicators remain comfortably below indicative thresholds in the baseline scenario (Figure 1 and Table 1). Under the baseline assumptions, the present value (PV) of public external debt expressed as a share of GDP increases slightly from 20 percent in 2014 to a peak of 23 percent in 2019, before gradually declining to about 19 percent by the end of the projection period. Debt service indicators are projected to rise gradually over time, reflecting the long-maturity nature of concessional loans which make up the majority of the existing external debt stock, as well as the increasing weight placed on new commercial borrowing.

14. The outlook for the external debt ratios remains favorable under standard stress tests (Figure 1 and Tables 3a-b). All debt burden indicators remain safely below their respective risk thresholds. However, stress tests show that external debt dynamics would be adversely affected in the event of a one-time 30 percent depreciation in the nominal exchange rate. The PV of external debt expressed as a share of exports is sensitive to a shock to financing terms, in which the interest rate on new public sector borrowing

[11] The figure is based on the government's estimation. Until June 30, 1999, the PSPF was a noncontributory pension scheme, with all benefits paid from the budget. The PSPF was transformed into a contributory defined benefit scheme on July 1, 1999, but benefits continued to be paid from the budget during a 5-year transition period (until June 30, 2004). From July 1, 2004 onwards, the PSPF has been paying pension benefits related to the pre-1999 scheme on behalf of the government, for which it has not been reimbursed.

[12] Correspondingly, for the purpose of this DSA, the fiscal accounts show an increase in domestic financing (below the line). As a result of this one-off stock adjustment, domestic debt reported in the DSA is higher than in staff projections.

increases by 200bps. In addition, the external debt outlook would become less favorable in the event of a permanent growth shock, whereby real GDP growth is lower at 5 percent (2 percentage points below the baseline).

15. Compared to the previous DSA, the favorable impact of the higher discount rate prevails over the adverse effects of changes to the projected fiscal deficit and financing assumptions. Slower fiscal consolidation leads to more debt accumulation, assumed to be financed with increasingly non-concessional external debt. (The assumed concessionality of new external borrowing continues to decline over the projection period, as can be seen by the declining grant element—Figure 1, top left panel). However, the increase in the discount rate to 5 percent reduces the present value of external PPG debt by about 2 percent of GDP on average.

PUBLIC DSA

16. Indicators of overall public debt (external plus domestic) and debt service do not point to significant vulnerabilities related to the level of domestic debt (Figure 2 and Table 2). In the baseline scenario, the PV of total public debt expressed as a share of GDP and revenues is expected to decline gradually over time. The PV of public debt peaks at about 36.3 percent of GDP in 2014, when various outstanding government liabilities are recognized, but remains well below the benchmark level of about 75 percent associated with heightened public debt vulnerabilities for strong performers.[13] The path of public debt is also consistent with the authorities' objective under the EAMU protocol of keeping the public debt-to-GDP ratio (in PV terms) below 50 percent.

17. Stress tests indicate that the sustainable public debt path hinges on continued fiscal consolidation. In a scenario assuming that the primary deficit (as a percent of GDP) remains at the projected 2014 level over the entire projection period, the PV of public debt would keep growing to 51.5 percent of GDP by 2034. This scenario highlights the critical importance of reducing fiscal deficits to prevent excessive debt buildup. Fiscal consolidation in the medium term may present challenges for the authorities in light of infrastructure needs and expenditure pressures to facilitate growth in key sectors (e.g., energy, transportation, and agriculture), as well as growing pension obligations and the upcoming general election in 2015. The most extreme shock corresponds to a 10 percent of GDP increase in debt-creating flows in 2015, which would capture some of the government implicit contingent liabilities and/or non-central government borrowing that is not included in the DSA. The amount of known contingent liabilities arising from pension obligations and government guarantees (about 9 percent of GDP) is broadly in line with the magnitude of shock assumed by the standardized stress test.

18. Growing pension obligations present a challenge to be met in order to preserve public debt sustainability. The DSA shows that recognizing the outstanding liability to PSPF has a noticeable impact on the public debt ratio. In the absence of a reimbursement for past benefit payments, PSPF is now in a

[13] See Revisiting the Debt Sustainability Framework for Low-Income Countries for a discussion of public debt benchmarks. Tanzania is still considered a strong performer as their CPIA score of 3.73 breaches the 3.75 threshold slightly for the first time. The public debt benchmark for medium performers is 56 percent of GDP.

fragile financial position. Moreover, pension obligations related to the pre-1999 scheme will continue to accrue and are ultimately a spending obligation of the government. More broadly, the actuarial deficit across the pension sector (consisting of several pension funds) gives rise to further liabilities. This highlights the need for the government to address its outstanding pension liabilities, budget for future payments related to the pre-1999 scheme, and undertake parametric pension reforms to improve the funds' financial positions.

POTENTIAL IMPACT OF NATURAL GAS RESOURCES

19. The baseline macroeconomic framework underlying the current DSA does not yet factor in the potential impact of possible future natural gas production from emerging offshore large-scale projects. Recent deep water exploration by major petroleum companies has confirmed large natural gas deposits. Public information released by the companies exploring for natural gas indicates that there are recoverable gas resources of at least 23-26 trillion cubic feet (TCF), although there remains significant uncertainty about exact reserve estimates. There is potential for the reserve estimates to increase further given the substantial exploration activities underway (although gas reserves are likely to remain significantly smaller than those of neighboring Mozambique). While exploration and appraisal are well underway, the final investment decisions to construct liquefied natural gas (LNG) terminals have not yet been made. (Such decisions are currently expected to be made in 2016.) Thereafter, the development phase would start, and it would take several years before commercial production and exports of LNG could begin. Similarly, significant fiscal revenues would not materialize for at least a decade from today.

20. If natural gas prospects were to materialize, the impact on debt sustainability would depend on a variety of factors. The development phase (until about 2020/21) would see a significant deterioration of the external current account, given the import-intensive nature of LNG projects. The extent of the deterioration depends on the scale of the projects, e.g., whether it would be a two-train or four-train LNG plant. The larger CA deficit would be financed by a combination of FDI inflows and public sector external borrowing depending on the government's decision on how to manage natural gas resources, i.e., whether gas-related investments would be led by the private or public sector.[14] A temporary increase in the fiscal deficit and public debt buildup is therefore possible in the medium term. Once LNG commercial production comes online (possibly in the early 2020s), natural gas exports would pick up significantly, alleviating pressures on the current account (by how much depends on the commercially viable amount, the portion earmarked for domestic consumption, and prevailing global natural gas prices— which are particularly volatile). The fiscal impact would build up gradually, with tax yields initially mitigated by deductions on investments. The overall long-term impact would likely be highly beneficial, with potentially large fiscal receipts from gas revenues (Box 2), thereby reducing public sector borrowing needs and enhancing debt sustainability. A more precise DSA scenario incorporating LNG developments will be

[14] The government's natural gas policy, which was adopted in October 2013, envisaged a more central role for the national oil and gas company, TPDC (Tanzania Petroleum Development Corporation), across the upstream, midstream, and downstream sectors. Tanzania is working to finalize gas-related policies and regulations, which will be encompassed in a new Gas Master Plan, the downstream-focused Natural Gas Act, and a new petroleum policy which will cover the upstream sector.

developed when commercially viable natural gas resources are known with greater confidence and when there is reasonable certainty that the development of a large scale gas project will go ahead.

Box 2. Offshore Natural Gas: Resources, Investment, and Fiscal Impact

Tanzania has made significant offshore gas discoveries during the last two years. The resource discoveries have been made by two groups of international petroleum companies that are still conducting exploration and appraisal; neither has yet declared the gas discoveries to be commercial. A final investment decision is expected at the earliest in late 2016, with production potentially commencing after 2020. The latest estimates indicate that there are total recoverable gas resources amounting to 23-26 trillion cubic feet (tcf). A rule of thumb is that 10-12 tcf of gas reserves is the minimum required for a two-train liquefied natural gas (LNG) plant.

This box presents estimates of the possible investment and fiscal impact of an LNG project applying a model developed for Tanzania by the IMF's Fiscal Affairs Department (FAD). With uncertainty about gas reserves, project cost and gas prices, the fiscal regime, and potential government equity participation, these are illustrative estimates, not forecasts.

Investment impact

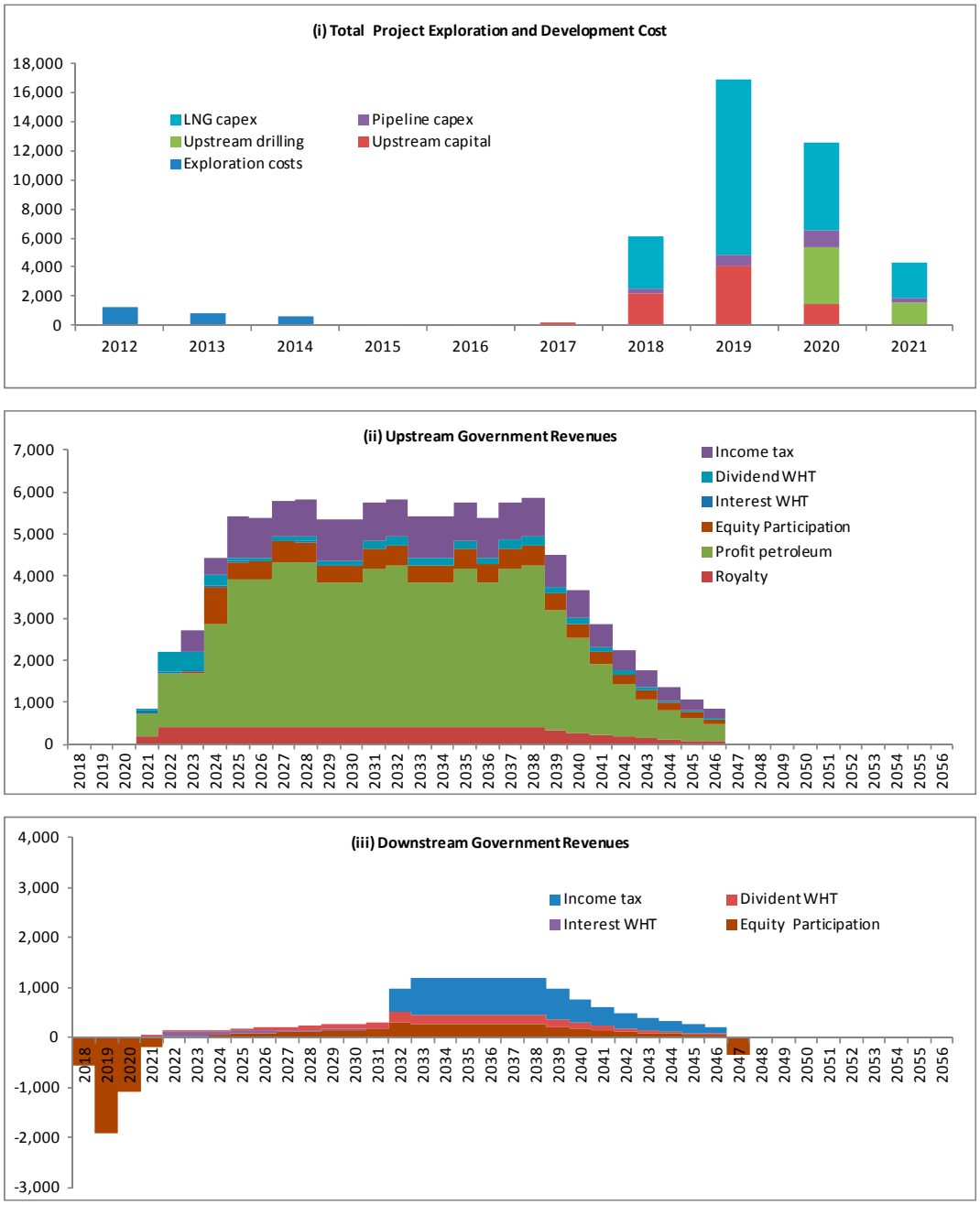
The investment impact of an LNG project will be significant, ranging from US$20-40 billion depending on the scale of the project.[1] Two-thirds of the development cost relate to pipelines and liquefaction facilities. During the development phase (possibly from 2018 to 2021), the sizeable investment will widen the current account deficit and temporarily boost growth. The fiscal impact during the development phase will depend on policy decisions regarding state participation in a project. As an illustration, if the government were to take a 15 percent equity share in the downstream of a large LNG project, the government's contribution to total development cost could be close to US$4.5 billion.

Fiscal revenue impact

Given the project uncertainty, and that the fiscal regime likely will evolve further, through either contractual negotiations or general reforms, it is only possible to simulate the fiscal impact under specific assumptions rather than make firm projections. The fiscal revenue impact would be highly beneficial. At full production, annual revenue collections by the government reach between US$3 billion and US$6 billion in the two different project simulations. These would have a transformational impact on the debt outlook, as by way of comparison today's public debt stock amounts to $12.4 billion. Most of the fiscal revenues would originate from the upstream, reflecting the higher effective tax rate and the assumption that the downstream will be regulated as a public utility. The fiscal regime is a hybrid combining royalty and income tax with production sharing. The largest revenue stream comes from production sharing, highlighting the importance of treating production sharing as any other source of fiscal revenue which should be paid to the Treasury and not retained by the national oil company.

[1] Simulations have been done for two project sizes: a two-train liquefaction plant with capacity for annual production of 10 million metric tons of LNG and a four-train plant producing 20 million metric tons per year.

Box Figure. Tanzania: LNG Gas Fiscal Simulations[1/]



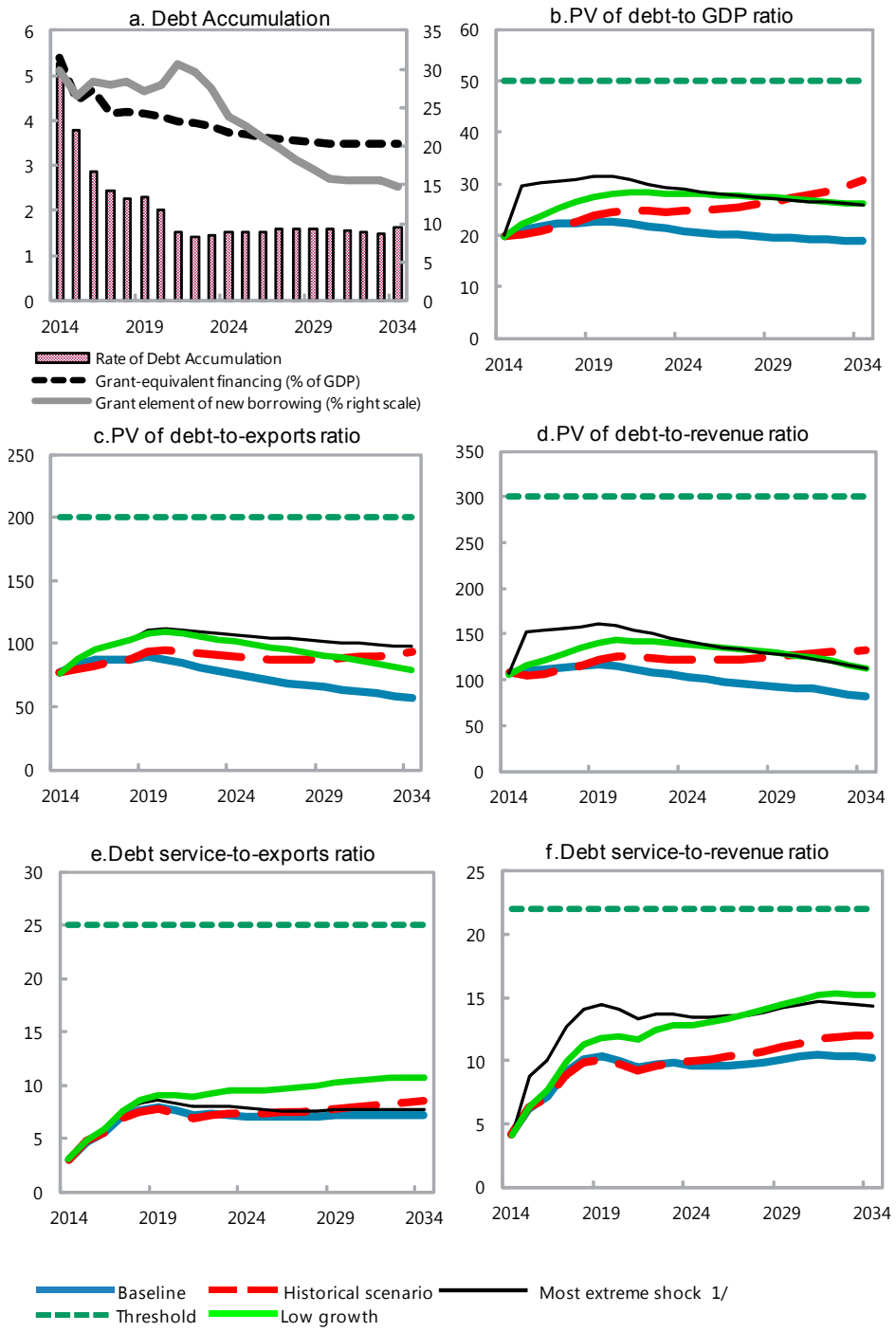
Source: IMF staff estimates

1/ Simulations of a four-train LNG plant with annual capacity of 20 mmt using total gas resources of 24 tcf over the life of the project.

CONCLUSION

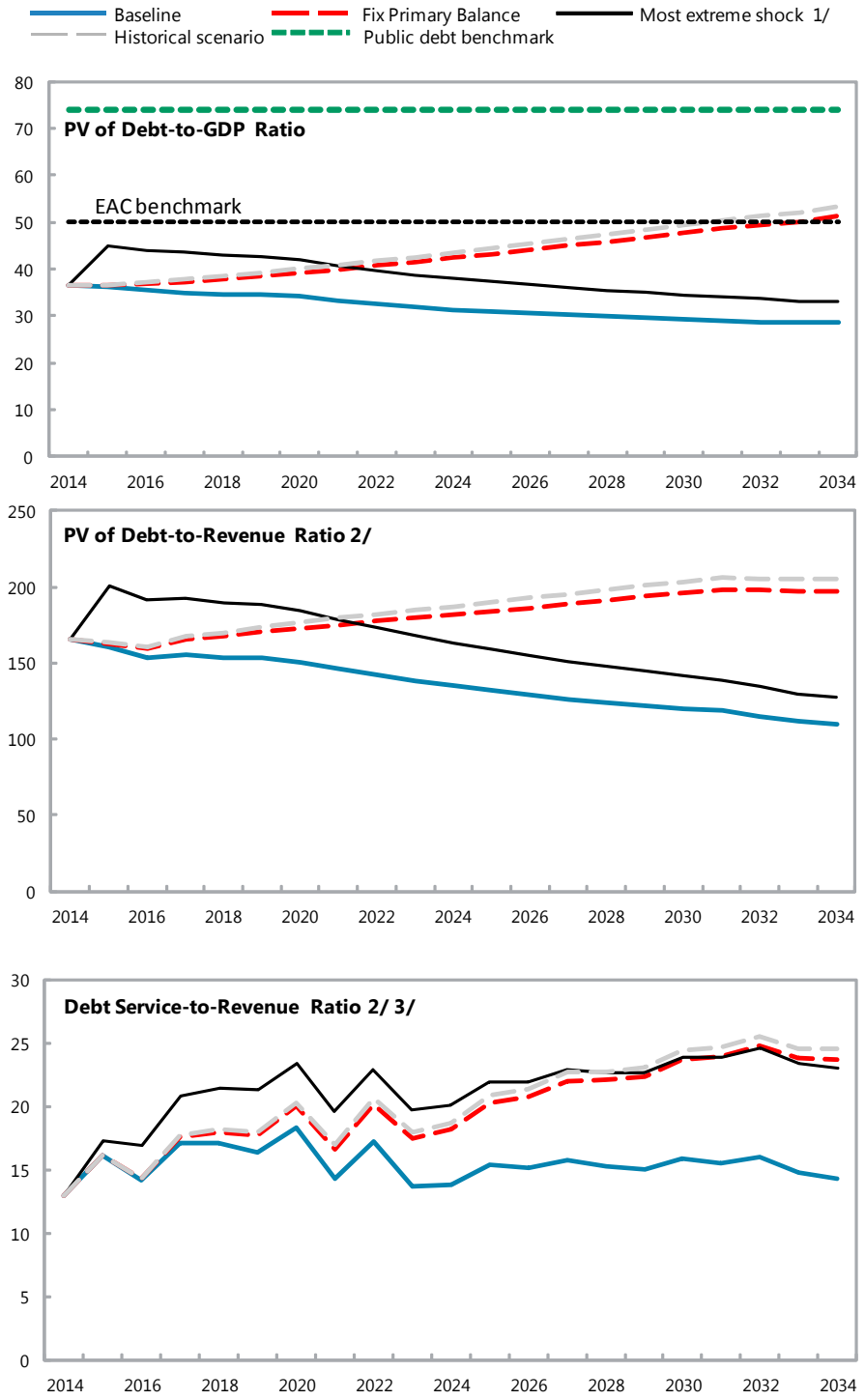
21. Tanzania continues to face a low risk of external debt distress and low risks from domestic public debt, but continued prudent policies are needed to preserve debt sustainability. Compared with the June 2012 DSA, the higher discount rate has a positive impact on the debt outlook, although this is partially offset by the assumption of a slower pace of fiscal deficit reduction and the assumed gradual reduction in the concessionality of external borrowing. While the public sector debt outlook remains favorable in the baseline, the public DSA, which only covers central government debt, shows that Tanzania's public debt would not stabilize in the absence of continued fiscal consolidation. The rising share of non-concessional borrowing would increase the average interest rate as well as potential exposure to interest rate fluctuations (if it took the form of variable rate loans) and rollover risks (if it took the form of Eurobond issuance with a lumpy repayment profile). Another challenge stems from government obligations related to the civil service pension sector, as well as yet-to-be-captured borrowing by local governments and public enterprises. Moreover, adverse shocks to economic growth, financing terms, or the exchange rate would significantly worsen the debt dynamics. A major upside risk relates to potentially large revenues from natural gas finds, which however would materialize only a decade from now. These results highlight the need for Tanzania to sustain fiscal consolidation efforts, to adopt a prudent approach to non-concessional borrowing, to initiate pension reforms, and to take concrete steps in improving its debt and public investment management capacity.

Figure 1. Tanzania: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2014–2034[1/]



Sources: Tanzanian authorities; and IMF staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation

Figure 2. Tanzania: Indicators of Public Debt Under Alternative Scenarios, 2014-2034[1]



Sources: Tanzanian authorities; and IMF staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.
3/ The volatile profile of debt service is due to the projected armotization for public domest

Table 1. Tanzania: External Debt Sustainability Framework, Baseline Scenario, 2011–2034[1]

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**35.1**	**34.0**	**34.3**			**36.8**	**37.8**	**38.0**	**38.3**	**38.5**	**38.7**		**35.7**	**31.0**	
of which: public and publicly guaranteed (PPG)	29.8	28.8	29.2			32.0	33.0	33.3	33.6	33.7	33.9		30.7	25.6	
Change in external debt	5.6	-1.1	0.2			2.6	0.9	0.2	0.3	0.2	0.2		-0.7	-0.2	
Identified net debt-creating flows	4.1	9.4	2.7			6.3	4.4	3.3	2.4	2.2	1.8		2.2	1.1	
Non-interest current account deficit	**9.1**	**18.1**	**13.7**	**9.5**	**4.4**	**13.9**	**12.8**	**11.7**	**11.1**	**10.6**	**10.1**		**9.8**	**8.2**	9.3
Deficit in balance of goods and services	12.4	19.2	14.8			15.1	14.0	12.9	11.9	11.4	10.8		10.0	8.0	
Exports	29.8	31.0	27.2			25.6	25.5	25.1	25.4	25.7	25.4		27.7	32.8	
Imports	42.2	50.2	42.1			40.8	39.5	38.0	37.3	37.1	36.2		37.7	40.8	
Net current transfers (negative = inflow)	-4.2	-3.6	-2.6	-3.9	0.6	-2.2	-2.0	-1.9	-1.7	-1.6	-1.5		-1.1	-0.5	-0.9
of which: official	-3.1	-2.4	-1.7			-1.2	-1.0	-0.9	-0.9	-0.8	-0.8		-0.5	-0.2	
Other current account flows (negative = net inflow)	0.9	2.5	1.4			1.0	0.8	0.6	0.8	0.8	0.8		0.8	0.7	
Net FDI (negative = inflow)	**-4.3**	**-6.1**	**-6.0**	**-4.7**	**1.0**	**-6.0**	**-6.7**	**-6.7**	**-7.0**	**-6.8**	**-6.7**		**-6.0**	**-6.0**	-6.1
Endogenous debt dynamics 2/	**-0.8**	**-2.6**	**-5.0**			**-1.6**	**-1.7**	**-1.7**	**-1.7**	**-1.7**	**-1.6**		**-1.5**	**-1.2**	
Contribution from nominal interest rate	0.2	0.3	0.4			0.5	0.6	0.7	0.7	0.8	0.8		0.8	0.8	
Contribution from real GDP growth	-1.9	-2.2	-2.0			-2.1	-2.3	-2.4	-2.4	-2.4	-2.4		-2.3	-2.0	
Contribution from price and exchange rate changes	0.9	-0.7	-3.3			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.5**	**-10.5**	**-2.5**			**-3.7**	**-3.5**	**-3.0**	**-2.1**	**-2.0**	**-1.5**		**-2.9**	**-1.2**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	22.2			24.7	25.9	26.4	26.8	27.1	27.4		25.8	24.2	
In percent of exports	...	...	81.7			96.2	101.5	105.2	105.7	105.3	108.1		93.3	73.8	
PV of PPG external debt	**...**	**...**	**17.2**			**19.9**	**21.2**	**21.7**	**22.1**	**22.4**	**22.6**		**20.9**	**18.8**	
In percent of exports	**...**	**...**	**63.2**			**77.5**	**83.0**	**86.5**	**87.2**	**86.9**	**89.3**		**75.4**	**57.4**	
In percent of government revenues	**...**	**...**	**98.5**			**107.1**	**109.7**	**111.1**	**113.2**	**114.4**	**115.9**		**102.8**	**81.8**	
Debt service-to-exports ratio (in percent)	**4.0**	**2.4**	**3.0**			**4.6**	**6.3**	**7.2**	**8.6**	**9.2**	**9.6**		**8.6**	**8.6**	
PPG debt service-to-exports ratio (in percent)	**1.0**	**1.3**	**1.9**			**3.0**	**4.7**	**5.6**	**7.0**	**7.6**	**7.9**		**7.1**	**7.2**	
PPG debt service-to-revenue ratio (in percent)	**1.9**	**2.3**	**3.0**			**4.1**	**6.2**	**7.2**	**9.0**	**10.0**	**10.3**		**9.6**	**10.2**	
Total gross financing need (Billions of U.S. dollars)	1.4	3.3	2.6			3.2	3.0	2.9	2.9	3.2	3.3		5.3	10.4	
Non-interest current account deficit that stabilizes debt ratio	3.5	19.2	13.4			11.4	11.9	11.4	10.8	10.4	9.9		10.5	8.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.7	6.7	6.9	7.0	0.3	7.1	7.1	7.1	7.0	6.9	6.9	7.0	6.9	7.0	6.9
GDP deflator in US dollar terms (change in percent)	-3.0	2.2	10.8	3.4	6.2	5.8	4.2	3.9	2.5	2.1	2.1	3.4	2.1	2.0	2.1
Effective interest rate (percent) 5/	0.9	0.9	1.3	0.9	0.2	1.7	1.9	2.0	2.1	2.2	2.2	2.0	2.3	3.0	2.5
Growth of exports of G&S (US dollar terms, in percent)	22.8	13.3	4.3	15.4	7.8	6.7	11.0	9.3	10.9	10.7	7.5	9.4	11.0	11.0	11.0
Growth of imports of G&S (US dollar terms, in percent)	19.5	29.5	-0.6	19.5	12.1	9.8	8.1	7.0	7.5	8.7	6.3	7.9	10.0	10.0	10.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	29.9	26.6	28.4	28.0	28.4	27.1	28.1	23.9	14.8	21.0
Government revenues (excluding grants, in percent of GDP)	16.4	17.6	17.5			18.6	19.3	19.5	19.5	19.5	19.5		20.3	23.0	21.0
Aid flows (in Billions of US dollars) 7/	1.1	1.2	1.0			1.9	1.8	2.2	2.2	2.4	2.6		3.5	7.6	
of which: Grants	1.1	1.2	1.0			1.2	1.2	1.5	1.4	1.5	1.7		2.6	6.2	
of which: Concessional loans	0.0	0.0	0.0			0.7	0.6	0.7	0.7	0.8	0.9		0.9	1.4	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.4	4.5	4.7	4.2	4.2	4.2		3.7	3.5	3.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			54.1	54.5	60.8	58.0	58.1	57.1		60.9	53.8	58.6
Memorandum items:															
Nominal GDP (Billions of US dollars)	23.7	25.8	30.6			34.7	38.7	43.0	47.2	51.5	56.2		86.7	207.8	
Nominal dollar GDP growth	3.6	9.0	18.6			13.3	11.6	11.2	9.7	9.1	9.1	10.7	9.1	9.1	9.1
PV of PPG external debt (in Billions of US dollars)			5.2			6.8	8.1	9.2	10.3	11.3	12.5		17.8	38.5	
(PVt-PVt-1)/GDPt-1 (in percent)						5.3	3.8	2.9	2.5	2.3	2.3	3.2	1.5	1.6	1.6
Gross workers' remittances (Billions of US dollars)	0.0	0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.1	0.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	17.2			19.9	21.1	21.7	22.1	22.3	22.6		20.8	18.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	62.9			77.2	82.7	86.2	86.9	86.6	89.0		75.2	57.3	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	1.9			3.0	4.7	5.6	6.9	7.6	7.9		7.0	7.1	

Sources: Tanzanian authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Tanzania: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–2034

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	40.4	39.8	41.3			48.7	47.9	47.0	46.5	46.0	45.8		41.2	35.3	
of which: foreign-currency denominated	29.8	28.8	29.2			32.0	33.0	33.3	33.6	33.7	33.9		30.7	25.6	
Change in public sector debt	6.6	-0.6	1.4			7.4	-0.8	-1.0	-0.5	-0.5	-0.1		-0.9	-0.1	
Identified debt-creating flows	5.1	-1.1	1.9			7.3	-0.1	-1.0	-0.4	-0.2	-0.1		-0.8	-0.2	
Primary deficit	5.6	4.0	5.2	3.7	1.6	3.6	3.2	2.4	2.5	2.6	2.6	2.8	1.7	1.6	1.7
Revenue and grants	21.1	22.1	20.8			22.0	22.4	23.0	22.5	22.5	22.5		23.3	26.0	
of which: grants	4.7	4.5	3.4			3.4	3.1	3.5	3.0	3.0	3.0		3.0	3.0	
Primary (noninterest) expenditure	26.7	26.0	26.0			25.6	25.6	25.5	25.0	25.1	25.1		25.0	27.5	
Automatic debt dynamics	-0.5	-5.1	-3.8			-3.4	-3.8	-3.4	-2.9	-2.8	-2.7		-2.5	-1.8	
Contribution from interest rate/growth differential	-2.2	-2.9	-2.3			-2.3	-3.0	-3.0	-2.8	-2.8	-2.7		-2.5	-1.8	
of which: contribution from average real interest rate	-0.1	-0.4	0.3			0.5	0.2	0.2	0.2	0.2	0.2		0.3	0.5	
of which: contribution from real GDP growth	-2.1	-2.5	-2.6			-2.7	-3.2	-3.2	-3.1	-3.0	-3.0		-2.7	-2.3	
Contribution from real exchange rate depreciation	1.7	-2.2	-1.5			-1.1	-0.8	-0.4	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.6			7.1	0.5	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			5.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.6			1.6	0.5	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.5	0.6	-0.5			0.2	-0.7	0.0	-0.1	-0.2	0.0		-0.1	0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	29.2			36.5	36.1	35.4	35.0	34.6	34.6		31.4	28.6	
of which: foreign-currency denominated	...	...	17.2			19.9	21.2	21.7	22.1	22.4	22.6		20.9	18.8	
of which: external	...	...	17.2			19.9	21.2	21.7	22.1	22.4	22.6		20.9	18.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.1	6.5	7.5			6.4	6.8	5.7	6.4	6.4	6.3		4.9	5.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	140.3			166.0	161.0	153.6	155.3	153.6	153.3		134.7	110.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	167.5			196.8	186.9	181.1	179.1	177.1	176.9		154.5	124.2	
of which: external 3/	...	...	98.5			107.1	109.7	111.1	113.2	114.4	115.9		102.8	81.8	
Debt service-to-revenue and grants ratio (in percent) 4/	6.9	11.3	10.9			13.0	16.1	14.2	17.1	17.0	16.4		13.8	14.3	
Debt service-to-revenue ratio (in percent) 4/	8.9	14.2	13.0			15.4	18.7	16.7	19.7	19.6	18.9		15.8	16.2	
Primary deficit that stabilizes the debt-to-GDP ratio	-1.0	4.5	3.8			-3.9	4.0	3.4	3.0	3.0	2.7		2.6	1.6	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.7	6.7	6.9	7.0	0.3	7.1	7.1	7.1	7.0	6.9	6.9	7.0	6.9	7.0	6.9
Average nominal interest rate on forex debt (in percent)	0.9	0.9	1.5	0.8	0.3	1.7	1.9	2.1	2.2	2.2	2.3	2.0	2.3	3.2	2.6
Average real interest rate on domestic debt (in percent)	2.1	-1.0	2.8	2.6	2.2	4.0	1.4	1.4	1.7	1.6	1.5	1.9	1.8	3.0	2.2
Real exchange rate depreciation (in percent, + indicates depreciation	7.7	-8.0	-5.7	-1.2	6.7	-4.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.1	10.4	9.9	8.1	1.5	7.3	6.3	5.9	5.3	5.1	5.1	5.8	5.1	5.0	5.1
Growth of real primary spending (deflated by GDP deflator, in percer	0.1	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	29.9	26.6	28.4	28.0	28.4	27.1	28.1	23.9	14.8	...

Sources: Tanzanian authorities; and IMF staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3a. Tanzania: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	20	21	22	22	22	23	**21**	19
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	20	20	21	22	23	24	**25**	31
A2. New public sector loans on less favorable terms in 2014-2034 2	20	22	24	25	27	28	**30**	32
A3. Alternative Scenario : Low growth	20	22	24	25	26	27	**28**	26
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	20	21	22	22	22	22	**21**	19
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	20	22	23	23	24	24	**21**	19
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	20	22	25	25	25	26	**23**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	20	23	24	24	25	25	**22**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	20	23	25	25	25	26	**23**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	20	29	30	31	31	31	**29**	26
PV of debt-to-exports ratio								
Baseline	78	83	87	87	87	89	**75**	57
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	78	79	83	85	88	94	**89**	93
A2. New public sector loans on less favorable terms in 2014-2034 2	78	87	95	100	103	110	**107**	97
A3. Alternative Scenario : Low growth	77	87	95	99	102	108	**101**	79
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	78	82	86	86	86	88	**74**	56
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	78	88	97	97	96	98	**81**	60
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	78	82	86	86	86	88	**74**	56
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	78	88	97	97	96	98	**80**	57
B5. Combination of B1-B4 using one-half standard deviation shocks	78	86	90	90	89	91	**75**	56
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	78	82	86	86	86	88	**74**	56
PV of debt-to-revenue ratio								
Baseline	107	110	111	113	114	116	**103**	82
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	107	104	106	111	115	121	**122**	133
A2. New public sector loans on less favorable terms in 2014-2034 2	107	115	122	129	136	143	**146**	139
A3. Alternative Scenario : Low growth	106	115	122	129	135	140	**138**	113
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	107	109	111	112	114	115	**102**	81
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	107	112	118	120	120	122	**105**	81
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	107	117	126	128	129	131	**116**	92
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	107	117	124	125	126	127	**109**	82
B5. Combination of B1-B4 using one-half standard deviation shocks	107	119	128	129	130	131	**114**	88
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	107	153	154	157	158	160	**142**	113

Table 3b. Tanzania: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2034 (concluded)

(In percent)

	Projectins							
	2014	2015	2016	2017	2018	2019	2014	2034
Debt service-to-exports ratio								
Baseline	3	5	6	7	8	8	**7**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	3	5	5	7	7	8	**7**	8
A2. New public sector loans on less favorable terms in 2014-2034 2	3	5	5	6	6	6	**7**	9
A3. Alternative Scenario : Low growth	3	5	6	8	9	9	**9**	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	3	5	6	7	8	8	**7**	7
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	3	5	6	8	8	9	**8**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	3	5	6	7	8	8	**7**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	3	5	6	7	8	8	**8**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	3	5	6	7	8	8	**7**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	3	5	6	7	8	8	**7**	7
Debt service-to-revenue ratio								
Baseline	4	6	7	9	10	10	**10**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	6	7	9	10	10	**10**	12
A2. New public sector loans on less favorable terms in 2014-2034 2	4	6	6	8	8	8	**9**	13
A3. Alternative Scenario : Low growth	4	6	8	10	11	12	**13**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	6	7	9	10	10	**10**	10
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	6	7	9	10	11	**10**	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	7	8	10	11	12	**11**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	6	7	10	11	11	**10**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	4	6	8	10	11	11	**11**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	4	9	10	13	14	14	**13**	14
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	13	13	13	13	13	13	**13**	13

Sources: Tanzania authorities; and IMF staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Tanzania: Sensitivity Analysis for Key Indicators of Public Debt 2014-2034

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	37	36	35	35	35	35	31	29
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	37	37	37	38	38	39	44	53
A2. Primary balance is unchanged from 2014	37	36	37	37	38	38	42	51
A3. Permanently lower GDP growth 1/	37	36	35	35	35	35	32	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	37	36	36	36	35	35	33	31
B2. Primary balance is at historical average minus one standard deviations in 2015-201	37	38	40	39	39	39	35	31
B3. Combination of B1-B2 using one half standard deviation shocks	37	37	39	38	38	38	34	31
B4. One-time 30 percent real depreciation in 2015	37	44	42	41	40	40	37	36
B5. 10 percent of GDP increase in other debt-creating flows in 2015	37	45	44	43	43	43	38	33
PV of Debt-to-Revenue Ratio 2/								
Baseline	166	161	154	155	154	153	135	110
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	166	163	161	167	170	174	187	206
A2. Primary balance is unchanged from 2014	166	162	159	165	167	170	182	197
A3. Permanently lower GDP growth 1/	166	161	154	156	154	155	138	120
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	166	162	156	158	157	157	141	119
B2. Primary balance is at historical average minus one standard deviations in 2015-201	166	170	173	175	172	171	149	119
B3. Combination of B1-B2 using one half standard deviation shocks	166	167	167	169	168	167	148	121
B4. One-time 30 percent real depreciation in 2015	166	195	183	183	179	177	157	139
B5. 10 percent of GDP increase in other debt-creating flows in 2015	166	201	191	193	190	189	163	127
Debt Service-to-Revenue Ratio 2/								
Baseline	13	16	14	17	17	16	14	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	16	14	18	18	18	19	25
A2. Primary balance is unchanged from 2014	13	16	14	18	18	18	18	24
A3. Permanently lower GDP growth 1/	13	16	14	17	17	16	14	15
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	13	16	14	17	17	17	14	15
B2. Primary balance is at historical average minus one standard deviations in 2015-201	13	16	15	19	19	18	15	15
B3. Combination of B1-B2 using one half standard deviation shocks	13	16	15	18	19	18	15	16
B4. One-time 30 percent real depreciation in 2015	13	17	17	21	21	21	20	23
B5. 10 percent of GDP increase in other debt-creating flows in 2015	13	16	17	22	21	20	17	17

Sources: Tanzanian authorities; and IMF staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.